A conserved comparison for fixed-charge memory returns

Force-defect inversion and strong finite-time bounds

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Abstract

The ordinary charged-tail estimate and the neutral Jacobi expansion give a direct fixed-sector treatment of the first memory return. An exact inverse of the body-loss identity bounds the force-sandwich defect at every fixed finite Schatten order. A finite-parameter energy function then approximates the scaled ordinary kernel at rate $O(p^{-1/2})$. The first return is a conserved background up to the same error; all later returns contribute $O(p^{-1})$. Exact energy pinching therefore reduces the total memory field to ordinary propagation in every fixed finite Schatten norm, uniformly on a compact central charge window and compact real-time intervals.

1 Model and result

Use standard CAR, $\mathcal{H}_m = \wedge^m \mathbb{C}^N$, $d_m = \binom{N}{m}$, and the common static Hermitian Gaussian model

$$H = \sum_{|I|=|J|=p} J_{IJ} c_I^\dagger c_J, \quad \mathbb{E} J_{IJ} J_{KL} = w_N \delta_{IL} \delta_{JK}, \quad w_N = \frac{4^p J_s^2}{\lambda_N} \left(\frac{p!}{N^p} \right)^2.$$

Overlaps are retained. Let $\lambda_N = p^2/N \to \lambda > 0$, $J_s > 0$ fixed, $\hbar = 1$, and use string time. Put

$$\|X\|_{r,m} = (\mathbb{E}\, \mathrm{tr}_m\, |X|^r)^{1/r}, \qquad C_m = w_N \binom{m}{p} \binom{N-m+p}{p}.$$

All bounds are uniform on $|m - N/2| \le L_0 p$ for fixed L_0, and on $|t| \le S$ for fixed S, with fixed $1 \le r < \infty$. The known fixed-charge setting is described in [7]. Write $h = H_m$ and define the exact dilation

$$V\psi = m^{-1/2} \sum_i |i\rangle c_i \psi, \quad K = I_N \otimes H_{m-1}, \quad Q = I - VV^\dagger,$$
$$a = V^\dagger K V = \alpha h, \quad \alpha = 1 - p/m, \quad R = QKV, \quad D_\perp = QKQ,$$
$$M(t) = R^\dagger e^{itD_\perp} R, \qquad \Lambda(t) = R^\dagger e^{itK} R.$$

The energy projection $\mathcal{P}_H X = \sum_e P_e X P_e$ retains entire degenerate blocks. Let

$$\mathcal{F}(t) = (1 - \mathcal{P}_H) m e^{-iht/2} M(t) e^{-iht/2}, \qquad X_0(t) = (1 - \mathcal{P}_H) m e^{-iht/2} \Lambda(t) e^{-iht/2}.$$

Theorem 1. *In the stated model, norms and limits,* $\sup_{m,t} \|\mathcal{F}(t) - X_0(t)\|_{r,m} = O(p^{-1/2})$.

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2 An exact inverse of the ordinary body-loss identity

On full neutral block families let

$$(DX)_m = mX_m - \sum_i c_i^\dagger X_{m-1} c_i, \qquad K_{n,m} = [p\mathcal{N} - D]H^n\big|_m.$$

Here $\mathcal{N}$ is Gaussian coupling number. Define the random force map and deterministic covariance map with their different inputs:

$$\begin{aligned} \Gamma_{H,m}(X_{m-1}) &= \sum_i [H, c_i]^\dagger X_{m-1}[H, c_i], \\ \Phi_m(Y_m) &= \mathbb{E}_{H'} H'_m Y_m H'_m, \\ F_{n,m} &= \Gamma_{H,m}(H^n_{m-1}) - 2p\Phi_m(h^n). \end{aligned}$$

The independent H' only defines Φ; all actual fields use the same original disorder. The established body product rule and Gaussian Euler–Laplacian identity [3, 1] give

$$K_{n+2,m} = \sum_{u+v+w=n} h^u F_{v,m} h^w, \qquad K_{0,m} = K_{1,m} = 0. \tag{1}$$

Indeed, both DH^n and $p\mathcal{N}H^n$ start with pnH^n; their second-order terms are respectively Γ_H and $2p\Phi$.

Lemma 1. *For every $n \geq 0$ the exact inverse is*

$$F_{n,m} = K_{n+2,m} - hK_{n+1,m} - K_{n+1,m}h + hK_{n,m}h. \tag{2}$$

Moreover $\|F_{n,m}\|_{r,m} \leq A_r(n+1)^{b_r} L_r^n \sqrt{n!}$ with constants independent of N, m, n on the stated window.

Proof. In (1), subtracting hK_{n+1} leaves only $u = 0$. Subtracting $K_{n+1}h - hK_nh$ then removes all $w \geq 1$. The argument includes $n = 0, 1$ by $K_0 = K_1 = 0$ and never commutes a force defect through h.

Equation (O) of [1] bounds K_n itself by a Gaussian factorial envelope. Alternatively, its main theorem bounds $K_n - \lambda_N \mathcal{N}(\mathcal{N}-1)H^n$, with $C_m^{-n/2}$ applied after loss. The OU/Chebyshev multiplier bound [3] gives $\|\mathcal{N}(\mathcal{N}-1)H^n\|_{r,m} \leq cn^2(1+\sqrt{2})^n \|H^n\|_{r,m}$. Together with $C_m \leq C_*$ and $\|H^n\|_{r,m} \leq L_r^n\sqrt{n!}$ this gives the same bound on K_n. In (2), use $2r$ Holder for each one-sided product and $4r$ for hK_nh. The extra factor $\sqrt{(n+2)!/n!}$ is polynomial. Only a fixed multiple of r is needed, independent of n. □

Thus the entire exponential series is summable in these norms:

$$F_m(t) = \Gamma_{H,m}(e^{itH_{m-1}}) - 2p\Phi_m(e^{ith}), \qquad \sup_{m,|t|\leq S} \|F_m(t)\|_{r,m} \leq A_{r,S}. \tag{3}$$

The tail is bounded by $A_r \sum_{n>L}(n+1)^{b_r}(L_rS)^n/\sqrt{n!}$. Finite-N identification with the displayed maps follows first by the exponential Taylor series; Gaussian moments control its products with the two linear force factors.

3 A finite-parameter conserved covariance comparison

Retain the exact finite-N quantities from [2]:

$$\theta_{j,m} = \prod_{u=0}^{p-1} \left(1 - \frac{jp}{m-u}\right)_+ \left(1 - \frac{jp}{N-m+p-u}\right)_+,$$
$$Q_{0,m}(x) = 1, \quad Q_{1,m}(x) = x, \quad Q_{j+1,m}(x) = xQ_{j,m}(x) - b_{j,m}Q_{j-1,m}(x),$$
$$b_{j,m} = C_m \sum_{v<j} \theta_{v,m}, \qquad T_{j,m} = p(\Phi_m - C_m\theta_{j,m})Q_{j,m}(h).$$

On the central window $b_{j,m}$ are uniformly bounded. The exact entire expansion and saved bounds are

$$e^{ith} = \sum_{j\geq 0} c_{j,m}(t)Q_{j,m}(h),$$
$$|c_{j,m}(t)| \leq A_S L_S^j/j!, \qquad \|T_{j,m}\|_{r,m} \leq A_r(j+1)^{b_r} L_r^j \sqrt{j!}. \tag{4}$$

The polynomials $Q_j(h)$ have an analogous factorial envelope, by their bounded recurrence coefficients and Gaussian moments. Consequently each of the following series converges in every fixed finite norm, uniformly on compact real times:

$$f_m(t,h) = \sum_{j\geq 0} c_{j,m}(t)\theta_{j,m}Q_{j,m}(h), \qquad Z_m(t) = \sum_{j\geq 0} c_{j,m}(t)T_{j,m}. \tag{5}$$

The exact identity before pinching is

$$\Phi_m(e^{ith}) = C_m f_m(t,h) + p^{-1}Z_m(t). \tag{6}$$

For each sample the scalar Jacobi series is entire in h, so $f_m(t,h)$ is a function of h, including at degenerate eigenvalues. Also $f_m(-t,h) = f_m(t,h)^\dagger$.

The map Φ_m/C_m is unital, completely positive and trace preserving. It contracts Schatten norms, and maps a unitary to an operator of norm at most one. Since C_m is bounded away from zero by $C_- > 0$ on the window, (6) implies

$$\sup_{m,|t|\leq S} \|f_m(t,h)\|_{r,m} \leq 1 + \frac{A_{r,S}}{pC_-} = O_{r,S}(1). \tag{7}$$

No samplewise operator bound on f is required. Define

$$\beta_m = \frac{2p^2}{m}C_m, \qquad b_m(t,h) = \beta_m f_m(t,h). \tag{8}$$

Then $[b_m(t,h),h] = 0$ exactly and all its fixed finite norms are uniformly bounded, since $p^2/m = O(1)$.

This comparison uses finite coefficients rather than replacing them by q^j. The standard q-Gaussian Mehler comparison used in [5, 8] remains compatible with the limit, but is not an additional input to the strong estimate here.

4 Ordinary propagation: every fixed finite norm

Let $F_i = [H, c_i]$, $\mathcal{A}_m = \sum_i F_i^\dagger F_i|_m$, $D_F = KV - Vh$, and $G = R^\dagger R$. Exact identities give

$$\begin{aligned} D_F^\dagger D_F &= \mathcal{A}_m/m, & R &= D_F + \delta V h, \quad \delta = p/m, \\ D_F^\dagger e^{itK} D_F &= \Gamma_{H,m}(e^{itH_{m-1}})/m, & 0 \le G &= \mathcal{A}_m/m - \delta^2 h^2 \le \mathcal{A}_m/m. \end{aligned} \tag{9}$$

All rectangular norms use the same source dimension d_m, even on the larger dilation space. Thus there is no change of trace normalization when applying rectangular Holder.

The sectorwise Gaussian force moments of [4] are

$$\mathbb{E}\,\mathrm{tr}_m\,\mathcal{A}_m^k \le (2k-1)!!(N v_N)^k, \quad v_N \le \frac{4J_s^2}{\lambda_N}\frac{p}{N}e^{2\lambda_N}. \tag{10}$$

They follow from both force covariance bounds, Hermitian dilation, Gaussian integration by parts, and Minkowski over i; the force components need not be independent. Hence, using a larger fixed integer moment when necessary,

$$\|\mathcal{A}_m\|_{r,m} = O_r(p), \quad \|G\|_{r,m} = O_r(p/m) = O_r(p^{-1}), \tag{11}$$

$$\|D_F\|_{2r,m} = O_r(p^{-1/2}), \qquad \|\delta V h\|_{2r,m} = O_r(p^{-1}). \tag{12}$$

The G estimate uses eigenvalue monotonicity, not operator monotonicity of positive powers. Unitary propagation yields $\|\Lambda(t)\|_{r,m} \le \|G\|_{r,m}$.

Expanding $R = D_F + \delta V h$ and using (12) gives

$$\begin{aligned} &\left\| p\Lambda(t) - pD_F^\dagger e^{itK} D_F \right\|_{r,m} \\ &\quad \le p\big(2\,\|D_F\|_{2r,m}\,\|\delta V h\|_{2r,m} + \|\delta V h\|_{2r,m}^2\big) = O_r(p^{-1/2}). \end{aligned} \tag{13}$$

Combine (3), (6), (8) and (9) without changing the lower-charge input:

$$\frac{p}{m}\Gamma_{H,m}(e^{itH_{m-1}}) - b_m(t,h) = \frac{p}{m}F_m(t) + \frac{2p}{m}Z_m(t). \tag{14}$$

Both terms are $O_{r,S}(p^{-1})$, because $m \asymp p^2$. It follows that

$$\sup_{m,|t|\le S} \|p\Lambda(t) - b_m(t,h)\|_{r,m} = O_{r,S,L_0}(p^{-1/2}). \tag{15}$$

The rate holds at each fixed finite r directly; it is not an interpolation of an L^2 rate with an $O(p)$ coarse bound. It concerns a leading ordinary kernel. Multiplying (15) by m/p would not give a bounded dynamic field.

5 The first return and its exact remainder

Write $U(t) = e^{iat}$ and use time convolution preserving matrix order. The finite-dimensional block Duhamel identity from [4] is

$$\Lambda = M - M * U * \Lambda, \qquad M = \Lambda + J_1 + E_1,$$

where $J_1 = \Lambda * U * \Lambda$ and $E_1 = M * U * \Lambda * U * \Lambda$. The minus sign in the first identity comes from two transitions and $i^2 = -1$. Auxiliary-kernel removal is an established Mori method [9]; here only finite matrices and finite iteration are used.

For $s \geq 0$, let $\Delta_s = \{v, w \geq 0 : v + w \leq s\}$ and $u = s - v - w$. Then

$$J_1(s) = \int_{\Delta_s} \Lambda(v) U(u) \Lambda(w)\, dv\, dw. \tag{16}$$

Define the finite-parameter conserved reference

$$\mathcal{K}_m(s,h) = \frac{m}{p^2} e^{-ihs} \int_{\Delta_s} b_m(v,h) e^{iau} b_m(w,h)\, dv\, dw. \tag{17}$$

Every factor commutes with h. Each fixed finite norm of $\mathcal{K}_m$ is uniformly bounded by Holder and $m/p^2 = O(1)$. Subtract the integrands in (16) and (17) as

$$(p\Lambda(v) - b_m(v,h)) U(u) (p\Lambda(w)) + b_m(v,h) U(u) (p\Lambda(w) - b_m(w,h)).$$

Apply Holder at $2r$, (15), and the bounds on $p\Lambda, b_m$. The simplex has area $s^2/2$, so

$$\sup_{m, 0 \leq s \leq S} \left\| m e^{-ihs/2} J_1(s) e^{-ihs/2} - \mathcal{K}_m(s,h) \right\|_{r,m} = O(p^{-1/2}). \tag{18}$$

No operator-norm bound on the comparison function is used.

The integrand of E_1 contains six factors R or $R^\dagger$ and unitary propagators. Holder at $6r$ on the common source normalization, including Gaussian expectation, gives

$$\|E_1(s)\|_{r,m} \leq \frac{|s|^4}{4!} \|G\|_{3r,m}^3, \qquad \sup_{m, |s| \leq S} \|m E_1(s)\|_{r,m} = O(p^{-1}). \tag{19}$$

This is an exact finite-return remainder, not a time Taylor truncation. Reverse the simplex variables to get $J_1(-s) = J_1(s)^\dagger$ and define $\mathcal{K}_m(-s,h) = \mathcal{K}_m(s,h)^\dagger$. The same bounds hold on both time signs. Combining them yields

$$\sup_{m, |s| \leq S} \left\| m e^{-ihs/2} (M(s) - \Lambda(s)) e^{-ihs/2} - \mathcal{K}_m(s,h) \right\|_{r,m} = O(p^{-1/2}). \tag{20}$$

Finally $(1 - \mathcal{P}_H)\mathcal{K}_m = 0$ samplewise and $\|1 - \mathcal{P}_H\|_{r \to r} \leq 2$. This proves Theorem 1.

6 Preparations, calibration and the next interface

At half filling let P_O be any already controlled fixed Gaussian-degree selected polynomial preparation. Its finite-N orbit density from [6] satisfies

$$\mathbb{E}\langle P_O^\dagger Y P_O\rangle_O = E\,\mathrm{tr}_m(R_{P,N}Y), \qquad \sup_N \|R_{P,N}\|_{2,m} < \infty$$

for covariant integrable Y. Set $\mathcal{E} = \mathcal{F} - X_0$. It is covariant, including its exact energy pinching. Thus

$$\mathbb{E}\,\|\mathcal{E}(s)P_O|O\rangle\|^2 \le \|R_{P,N}\|_{2,m}\,\|\mathcal{E}(s)\|_{4,m}^2 = O(p^{-1}). \tag{21}$$

This is a strong return-error estimate on the fixed polynomial preparation core. It does not require interchanging exact pinching and the large-N limit. It also does not by itself bound arbitrary products of actual total-memory fields: their other factors still need uniform high moments.

The reference is nonzero before pinching. Since $f_m(0,h) = 1$ and $\partial_t f_m(0,h) = i\theta_{1,m}h$, (17) gives, at fixed N,

$$\begin{aligned}\mathcal{K}_m(s,h) = {}& \frac{2p^2}{m}C_m^2 s^2\, I \\ & + i\frac{2p^2}{3m}C_m^2(2\theta_{1,m}+\alpha-3)hs^3 + O(s^4).\end{aligned} \tag{22}$$

To obtain the cubic term use $\int_{\Delta_s} v = \int_{\Delta_s} w = \int_{\Delta_s} u = s^3/6$, and retain the outside e^{-ihs}. The fixed-order coefficients have bounded finite moments. At half filling, $p^2/m \to 2\lambda$, $C_m \to C = J_s^2/\lambda$, $\theta_{1,m} \to q = e^{-4\lambda}$, and $\alpha \to 1$. The first two coefficients therefore reproduce the existing background calibration of [5]:

$$4\lambda C^2 s^2 I + i\frac{8\lambda C^2}{3}(q-1)hs^3.$$

This consistency check is not a new finite-time Mehler formula.

The proved reduction is $\mathcal{F} - X_0 = O(p^{-1/2})$ in every fixed finite norm, in the actual fixed charge sectors. The next task is the fluctuation-scale two-component description of X_0. It still requires the central four-auxiliary-field conditional regression and the charged finite-α interface. The actual bilocal Volterra reconstruction must then be transferred with its required high moments. Arbitrary interlaced nonpolynomial selected blocks, the spatial/mass dictionary and finite-G gravitational dynamics are not conclusions of this note.

7 Saved finite checks and attribution

The independent finite checks retain all Gaussian chaoses and both adjacent charge sectors. They do not replace the analytic bounds or establish a double-scaling limit by numerical fitting.

Check	Parameters	Coverage
Free noncommutative inverse	$n = 0, \dots, 20$	21 exact integer rows
CAR/Hermite inverse	$(N, p, m) = (3, 1, 2)$	$n = 0, 1, 2, 3$
	$(4, 1, 2)$	$n = 0, 1, 2$
	$(4, 2, 3)$	$n = 0, 1$
Block propagation	same three systems	five real times each
Return norm inequalities	$r = 1, 2, 4$	90 inequalities

All nine CAR rows have a nonzero lower-chaos part. The real Gaussian coordinate basis is normalized by $C_m = 1$, with the same coordinates in charge m and m-1. The block checks use seeds 731, 732, 733 and times $-1.1, -0.2, 0, 0.4, 1.1$. Upper-triangular block matrix exponentials evaluate the ordered convolutions directly, with no quadrature or time cutoff. They check $M = \Lambda + J_1 + E_1$, the force-dilation identity, and the source-normalized bounds on J_1, E_1. Floating residuals and tolerances are retained in the JSON; they are not interval certificates. The formal free-word cancellation uses exact integer arithmetic.

The body-loss and Gaussian product identities, the ordinary factorial tail, the neutral Jacobi decomposition and the finite Volterra expansion are inherited results. This note combines them through the exact inverse and a finite-parameter conserved comparison to obtain the sectorwise strong return estimate. No priority claim is made for the elementary inverse or the standard operator inequalities. The local evidence record binds the code, scientific output, proof inputs, compilation and pages.

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